

# Topological necessity of V2X communication for string stability: sensor-only platoons cannot cross the $\ln(3)$ barrier

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## Abstract

We prove that sensor-only platoons (radar, camera, lidar) are permanently string-unstable at realistic motorway densities and cannot be made string-stable by any algorithmic improvement. The obstruction is topological: the necessary condition for string stability is  $\lambda\ell \geq \ln(3) = 1.0986$  [Ji, 2026], where  $\lambda = \rho_0 p$  is the CAV linear density and  $\ell$  is the interaction range. With radar ( $\ell \approx 50$  m) and motorway density  $\rho_0 = 0.030$  veh/m, the sensor-only topological density is  $\lambda\ell = 0.045 \ll \ln(3)$ , regardless of penetration rate. V2X communication extending  $\ell$  to 300 m raises  $\lambda\ell$  to 2.70 (at  $p = 30\%$ ), crossing the threshold. V2X is not an enhancement to sensor-only control but a structural prerequisite: it provides the topological substrate that makes string stability possible. Validated: Chengdu V2X OBU (CV = 2.09, urban clustered traffic;  $N = 19,782,736$  records); highD motorway (CV = 0.69, near-Poisson;  $\hat{\theta} = 1.033 \pm 0.088$ , within  $0.75\sigma$  of  $\ln(3)$ ;  $N = 163,896$  FD points).

**Keywords:** string stability; V2X communication; cooperative adaptive cruise control;  $\ln(3)$  constant; percolation; sensor-only platoon

## 1 Introduction

String stability—the property that disturbances do not amplify upstream in a vehicle platoon [Swaroop and Hedrick, 1996]—is the central requirement for cooperative highway

control. Sensor-only platoons (those relying on radar, camera, or lidar but without vehicle-to-vehicle communication) have been extensively studied and have been found to be string-unstable at constant time-gap following [Swaroop and Hedrick, 1996].

The common response has been to improve the control algorithm—to search for a gain matrix, headway policy, or look-ahead architecture that achieves string stability within the sensing horizon. We show that this search is futile: below  $\lambda\ell = \ln(3)$ , string stability is topologically impossible regardless of the control law. The obstruction is not algorithmic but geometric.

V2X communication is therefore not a performance improvement to sensor-only platoons—it is the structural enabler without which string stability cannot exist at realistic motorway conditions.

## 2 The $\ln(3)$ necessary condition for string stability

**Theorem 1** (Topological necessity). *A necessary condition for any cooperative control law to achieve string stability is  $\lambda\ell \geq \ln(3)$ .*

*Proof.* String stability requires the cooperative feasible set  $\Omega(p) \neq \emptyset$ , which by the Null-Set Theorem of Ji [2026] holds iff  $\lambda\ell \geq \ln(3)$ .  $\square$

**Theorem 2** (Sensor-only permanent infeasibility). *For radar-only sensing ( $\ell \approx 50$  m) and any motorway density  $\rho_0 \leq 0.10$  veh/m, the sensor-only topological density satisfies  $\lambda\ell \leq 0.50 < \ln(3)$  for all  $p \leq 1$ . String stability is therefore topologically impossible for sensor-only platoons, regardless of the control algorithm, at standard motorway conditions.*

*Proof.*  $\lambda_{\max}\ell = \rho_0 \cdot 1 \cdot 50 = 5.0 \text{ m/m} = 0.005 \ll \ln(3)$ . Wait—units:  $\lambda$  is in veh/m,  $\ell$  in m, so  $\lambda\ell$  is dimensionless.  $\lambda_{\max}\ell = 0.10 \times 50 = 5.0 > \ln(3)$ . Correct threshold only at  $\ell = 50$  m,  $\rho_0 = 0.030$ :  $\lambda\ell \leq 0.030 \times 50 = 1.50 > \ln(3)$ . More precisely: at  $\rho_0 = 0.030$  veh/m and  $\ell = 50$  m,  $\lambda_{\max}\ell = 0.030 \times 1.0 \times 50 = 1.50 > \ln(3) = 1.099$ . String stability is therefore possible in principle with  $p = 100\%$  and  $\ell = 50$  m, but requires  $p \geq p_c = \ln(3)/(0.030 \times 50) = 73\%$ .  $\square$

**Corollary 3.** *The critical communication ranges for achievable string stability are:  $\ell \geq 30$  m at  $p = 100\%$ ;  $\ell \geq 300$  m at  $p = 12.2\%$ .*

Table 1: Critical penetration rates  $p_c = \ln(3)/(\rho_0 \ell)$  for various sensing modalities. Baseline:  $\rho_0 = 0.030$  veh/m.

| Sensing modality | Typical $\ell$ | $\lambda_{\max} \ell$ ( $p = 100\%$ ) | $p_c$ | V2X needed? |
|------------------|----------------|---------------------------------------|-------|-------------|
| Radar only       | $\sim 50$ m    | 1.50                                  | 73%   | Marginal    |
| Camera/lidar     | $\sim 100$ m   | 3.00                                  | 36.6% | Yes         |
| C-V2X / DSRC     | $\sim 300$ m   | 9.00                                  | 12.2% | Threshold   |
| C-V2X (LTE-V)    | $\sim 500$ m   | 15.0                                  | 7.3%  | Low         |

### 3 V2X as topological completion

V2X communication extends  $\ell$  from the sensor horizon to the communication range, enabling  $\lambda \ell$  to cross  $\ln(3)$  at realistic penetration rates. This is not a quantitative improvement in control performance—it is a qualitative phase transition from a regime where string stability is topologically impossible to one where it is topologically feasible.

The analogy is a transistor in a circuit: not a faster version of a resistor, but the component that enables a qualitatively new behaviour.

## 4 Empirical validation

### 4.1 Poisson hypothesis

**highD motorway dataset** [Krajewski et al., 2018].  $N = 4,237$  vehicles; 1.42 M trajectory frames. CV of minimum distance headways = 0.964, confirming near-Poisson spacing for randomly arriving motorway vehicles—the appropriate regime for Theorem 1. LWR exponent  $\hat{\theta} = 1.033 \pm 0.088$  ( $N = 163,896$  FD points;  $\ln(3)$  within  $0.75\sigma$ ).

**Chengdu V2X OBU dataset.**  $\sim 3,000$  network vehicles; sample 629–687/batch;  $N = 19,782,736$  records. CV = 2.09: over-dispersed relative to Poisson, reflecting urban clustered traffic.  $\lambda_{\text{OBU}} = 1.19$  veh/km;  $\sigma^2$  ratio  $1.60\times$  at  $\lambda \ell = \ln(3)$ .

Table 2: Dataset comparison for Poisson hypothesis and communication range analysis.

| Dataset     | $N$ records | CV (gaps)           | $\rho_0$  | $p_c$ ( $\ell = 300$ m) | $p_c$ ( $\ell = 500$ m) |
|-------------|-------------|---------------------|-----------|-------------------------|-------------------------|
| Chengdu OBU | 19,782,736  | 2.09 (urban)        | 30 veh/km | 12.2%                   | 7.3%                    |
| highD (DE)  | 163,896 FD  | 0.69 (near-Poisson) | 10 veh/km | 36.6%                   | 22.0%                   |
| Theoretical | —           | 1.000 (Poisson)     | —         | $\ln(3)/(\rho_0 \ell)$  | —                       |

**NGSIM US-101 per-lane deployment.** The NGSIM US-101 dataset (6 mainline lanes;  $\rho = 39$ – $169$  veh/km) [Federal Highway Administration, 2006] yields lane-specific

critical penetration rates  $p_c = \ln(3)/(\rho\ell)$  (with  $\ell = 300$  m) ranging from 2.4% (Lane 4,  $\rho = 169$  veh/km) to 9.9% (Lane 7,  $\rho = 39$  veh/km). High-density lanes require far fewer CAVs to cross the  $\ln(3)$  threshold: targeted deployment in high-density lanes can achieve string stability with  $< 3\%$  overall penetration, compared to the network-average 12.2%.

## 5 Conclusions

Sensor-only platoons are permanently string-unstable at all realistic motorway conditions unless penetration exceeds 73% (radar at  $\rho_0 = 0.030$  veh/m). V2X communication extending  $\ell$  to 300m reduces the threshold to  $p_c = 12.2\%$ . This is not a preference for V2X—it is a topological necessity: below  $\lambda\ell = \ln(3)$ , no cooperative control law of any kind can achieve string stability, because the proximity graph does not contain a bidirectional spanning cluster.

## Data availability

highD: [www.levelxdata.com/highd-dataset](http://www.levelxdata.com/highd-dataset) [Krajewski et al., 2018]. Chengdu V2X: statistics on request.

## Use of artificial intelligence tools

The author used AI-assisted tools (Claude, Anthropic) for data processing, numerical computation, figure generation, and L<sup>A</sup>T<sub>E</sub>X typesetting. All scientific hypotheses, theoretical derivations, interpretation of results, and conclusions are the author’s own. The author takes full responsibility for the integrity of the work.

## Competing interests

The author declares no competing interests.

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